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The design and evaluation of a picture fuzzy mining safety system with picture fuzzy differential equations

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ABSTRACT

The mining industry continues to play a significant role in developing the economy in developing countries. The need for minerals influences the magnitude of the mining industry, and safety concerns have taken on considerable importance in this sector. Equipment failure and breakdown will severely disrupt production and cause accidents for the mining workers. Most safety studies in the mining industry consider deterministic models with crisp parameters. This study addresses safety in the mining industry under uncertainty and fuzziness. We present a picture fuzzy mining safety system with picture fuzzy differential equations. We examine and discuss the solutions to a mining safety model using triangular picture fuzzy numbers. We further study the shutdown probability at the mining system in safe and unsafe modes.

1. Introduction

The mining industry continues to play an essential role to boost human civilization through social and economic development. In the current modern industrial era, with the increasing energy demand, the volume of the mining industry has increased in many folds and consequently, various safety issues in the mining industry have become very significant nowadays. Extraction of minerals is carried out mainly by three methods namely (i) underground mining, (ii) opencast mining, and (iii) underwater mining. The selection of the mining method is decided by the location of the mineral and the depth of the formation of the mineral in the subsurface. Minerals formed in the solid ground are extracted either by the underground or by the opencast system. Where the depth of the subsurface is less, it is taken by the opencast mining system. In the case of mining coal, the choice of the mining system is decided by the values of the stripping ratio (quantity of overburden removal in the cubic meter for extraction of one tonnage of coal production). Coal can be extracted by an opencast system when the stripping ratio value is less or equal to 10; otherwise, underground extraction system is practiced. The underwater is carried out for the precious mineral in the sea bed. The operation is also called offshore mining. It is to be noted that the extraction of petroleum is different from this system. In an underground system, the entry to the mineral is done in three ways namely (a) through the vertical shaft, (b) through the incline, and (c) through the adit. Entry to the shallow-depth minerals is practiced through the incline shaft method whereas the vertical shaft method is followed when minerals exist at a higher depth from the ground-surface. The minerals are deposited in the hilly terrain, and entry to the minerals is done by the adit method. On the other hand, in an opencast mining system, different benches are formed for the removal of overburden (waste rock) and extraction of minerals.

During the extraction of minerals from underground mines, the workers have to face various adverse situations. Several dangers are associated with mining minerals from subterranean systems of work which include runaway of tubs in haulages, snapping of winder ropes, explosions, roof fall, inundation, presence of intolerant level of poisonous gases, etc., in underground mines. Similar risks have been associated with the labor in the opencast system like the possibility of dumper runways, worksite bench slides, rock blasting strikes, etc. These issues not only harm the workers, but they can also be fatal in many cases. For all these reasons, the study of mining safety systems become critical and crucial. Researchers in the past and present are conducting and implementing various measures on safety issues in the mining industry. Li et al. [1] identified high-risk hazards and accident-prone factors in the coal industry. They identified six key risk variables for safety and related factors associated with coal mines through sensitivity and critical path analyses. Ali and Pal [2] highlighted the significant issues of both deadly and non-fatal injuries in the case of underground coal mining. They also investigated the relationship between coal miner 's safety behaviors and injuries and suggested various measures for better health and safety for the coal mine workers using artificial intelligence (AI) and virtual reality. Yang et al. [3] studied how coal mining disaster cover-ups were exposed as an alternate viewpoint to

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China's new safety narrative. Xue et al. [4] addressed the experimental determination of the energy strength threshold value for an outburst in coal mines for mining safety. Ma [5] highlighted the limitations of coal mining safety management efficiency. Wang and Li [6] explored the coal mining process's safety management. Yang et al. [7] addressed on mining worker safety and the use of information technology in coal mining. Prabu et al. [8] developed the smart coal mine monitoring safety system. Yang and He [9] focused on sustainability-based environmental risk and the coal-mine safety management index system. Danish and Onder [10] fruitfully utilized fuzzy logic to anticipate mine fires in underground coal mines. Shi et al. [11] developed fuzzy fault tree analysis for gas explosions in underground coal mines. Guo and Yang [12] described China's understanding of mining safety research trends and future needs for a sustainable mining industry. Bari et al. [13] described a hybrid multi-criteria decision-making technique for operational hazard assessments of heavy fuel oil-based power plants. Yinusa et al. [14] proposed using nonlinear modeling and analysis to remove atmospheric gaseous contaminants from cloud droplets. Abbasi and Choukolaei [15] researched the green supply chain network design technique through a literature analysis with an emphasis on carbon policy. Martínez-Arellano et al. [16] presented a data analytics methodology for better process control in flexible manufacturing cells. The existing literature reveals that the majority of research on mining system safety issues has been conducted in crisp environments and few of them addressed in fuzzy environments. However, the ordinary fuzzy environment is not robust enough to capture ambiguity and uncertainty involved in the mining safety system. In the ordinary fuzzy environment even including intuitionistic fuzzy concept, we can represent the degree of membership and non-membership only; but fails to provide any indication of rejection. However, the real world mining safety system the indication of rejection is common and crucially important factor. Thus, it is essential to study the mining safety system in picture fuzzy environment where there is flexibility of incorporation of the case of rejection in a suitable manner. So far, no study on the safety of mining systems has been done in the picture fuzzy environment. In this article, we mathematically formulated the Mining Safety Model System (MSMS) in terms of differential equations, which include two parameters namely the MSMS-safe-failure-rate and the MSMS-unsafe-failure-rate and three dependent variables that measure the probability of the MSMS operating normally, failing safely and failing unsafely. Here, from a practical and realistic point of view, we assume the model parameters are picture-fuzzy numbers. Thus, our model system turns into a picture fuzzy differential equation (PFDE).

In realistic scenarios, it is observed that the model's parameters are imprecise in nature. Thus, there is a need to handle the underlying mathematical model in an imprecise environment. Zadeh [17] developed the FS theory, which substantially unfolds the study of impreciseness in many directions. İç and Yurdakul [18] suggested a novel long-term (strategic) ranking methodology for machining center selection decisions based on a triangular fuzzy number assessment of machining center structural components. It is to be mentioned that ordinary fuzzy environments fail to accurately express rejection criteria. Thus, Coung and Kreinovich [19] introduced the concept of a picture fuzzy number that is suitable to incorporate various rejection concepts in an imprecise environment.

Many real-life problems of science & engineering have been mathematically formulated as differential equations, where the model parameters are assumed as accurate values although reality compels to be uncertain or imprecise. Thus, Fuzzy Differential Equation (FDE) [20,21] has been proposed to make these models are more realistic. Mondal and Roy [22] discussed the solution of a second-order differential equation in generalized trapezoidal fuzzy scenario. Eljaoui et al. [23] solved a second-order FDE with generalized Hukuhara differentiable using unsatisfactory and inconsistently approach. Vasavi et al. [24] used a variant of the constants formula under substantially generalized differentiable to present the principle of Newton's law of cooling in several forms of a first-order linear fuzzy difference equation. Mahata et al. [25] investigated the glucose-insulin regularity system in terms of FDE. Mondal et al. [26] introduced the various representation of nonlinear interval-valued fuzzy number and their application in difference equations. Mondal et al. [27] presented the adaptive strategies for the system of first-order FDEs and applied it into the arms race model. Devi and Ganesan [28] addressed an electric circuit problem in terms of FDE. Mondal et al. [29] addressed on using an FDE system to apply safety measures to the oil and gas sector. However, maximum of these works addressed the underlying FDE assuming the parameter values and/or initial condition(s) as simple fuzzy numbers with single membership value. Later on, the IFDE [30] was introduced with a degree of membership and non-membership function. Lei and Xu [31] introduced fundamental properties of IFS calculus and IFDE. Mondal and Roy [32,33] presented triangular intuitionistic fuzzy numbers as the initial value of a first-order, homogeneous and non-homogeneous differential equation. Ettoussi et al. [34] devised the sequential approximation approach for solving IFDE. Mondal and Roy [35] outlined the generalized intuitionistic fuzzy Laplace transformation's and fruitfully use in a electrical circuits problem. Mondal and Roy [36] addressed a first-order system of differential equations using a triangular fuzzy number as the initial value. Akin and Bayeg [37] studied the Zadeh extension principle-interpreted intuitionistic fuzzy initial value issue. Karar et al. [38] discussed about the best adaptive intuitionistic fuzzy logic control for a delivery system of anti-cancer drugs. Rani and Jamshida [39] utilized the Runge–kutta method to introduce a numerical solution of an IFDE. Tudu et al. [40] outlined a novel approach to solving type-2 fuzzy differential equations with an application to the arms race model. Melliani et al. [41] utilized the homotopy analysis approach to provide an approximate solution to the IFDE using the linear differential operator. But all these works did not consider refusal terms which are common in real-life problems. The concept of PFS was developed to incorporate such refusal situations of reality. Dhivya et al. [42] introduced first-order PFDE and illustrated a numerical example of drug distribution in the human body for different drug levels. In reality, in many situations, it was observed that the collected data are insufficient and transmit some misinformation. In such situations, incorporating the picture of fuzzy numbers offers better results. Thus, in this article, we have presented the mining safety model system [43] through picture fuzzy differential equations. We have discussed both analytically and numerically the solution of the mining safety model considering the underlying model parameters as triangular picture fuzzy numbers (See Table 1).

Table 1
List of abbreviations throughout the article.

Abbreviation	Description
CS	Crisp set
FS	Fuzzy set
IFS	Intuitionistic fuzzy set
CSNs	Crisp number
PFS	Picture fuzzy set
PFNs	Picture fuzzy number
TPFNs	Triangular picture fuzzy number
ODE	Ordinary differential equation
FDE	Fuzzy differential equation
IFDE	Intuitionistic fuzzy differential equation
PFDE	Picture fuzzy differential equation
IC	Initial condition
MSMS	Mining safety model system

1.1. List of abbreviation

1.2. Equipment used in mining system

Electrical substation with all accessories like a transformer, circuit breaker, switch banks, etc. common for both opencast and underground

mining systems. Some specific equipment related to underground and opencast mining systems is as follows:

In the underground mining system

• In the vertical shaft winder with all accessories and in the incline man-rider.

• Haulage system for mineral evacuation.

• Conveyor system for mineral evacuation.

• Loading at working face by SDL (side discharge loader), LHD (load haul dumper), and GAL (gathering arm loader).

• Mineral production by drilling and blasting. Drilling is done by handheld electric drill (coal mines) and jack hammer drill (metals).

• Roof support is done by twin bolter and quad bolter for roof bolting, hydraulic props, and powered support in longwall face (for coal mines).

• In metal mines rock bolting, cable bolting, etc. with the use of jackhammer drill.

In opencast mining system

• For loading shovels both operated electric and hydraulic.

• For transportation dumpers.

• For leveling dozer.

• For preparation of mineral different types of drill.

• Dragline in coal mines.

In most cases, it is observed that mathematical models are solved in crisp environments, which fails to depict the true risk of Mining Safety. Here, in this article, we have presented a mathematical model of Mining Safety Model Systems in picture fuzzy environment which evaluate the failure rate of safe mining and unsafe mining. We have expressed the mining safety model in terms of a picture fuzzy differential equation and discussed its solution in view of the probability of system failure in safe and unsafe modes.

1.3. Motivation

Studies on the picture fuzzy differential equation and its application in real-life scenarios are uncommon; according to a survey of the literature, no such research paper has been found that has applied the research work to any problem related to mining safety. On the other hand, the mining safety model is resolved by using the crisp number as the model's input. We note that the parameters of any model related to real problems do not indicate absolute values but rather approximate values. This leads to an adequate and consistent solution derived from these values. In these circumstances, picture fuzzy sets provide a superior outcome. However, we point out that the mining safety model must still be solved in a picture-fuzzy environment. We have noticed this research gap and are particularly motivated to carry out research work on this topic. In this article, we mathematically formulated a mining safety model and discussed analytically and numerically the solution of the model in a crisp and picture-fuzzy environment.

1.4. Novelty

Although some basic patterns of mining model system are followed, in this article some fresh attentions and new efforts have been made to study the mining safety system from realistic point of view. It is observed that there are certain kinds of inconsistencies in the outcomes of mining safety model system when it dealt in a crisp environment. In a mining safety system, the rejection criteria becomes obvious due to various constraints. Now in picture fuzzy environment there is an opportunity to incorporate rejection criteria in a suitable manner. Thus, there is a need to study the mining safety system in picture fuzzy environment. In this picture fuzzy environment, the rejection criteria are stated as positive membership, neutral membership and negative membership functions respectively. These three membership functions have the property lying between 0 to 1, and the sum of the three membership functions does not exceed 1. Till date, no study on mining safety systems has been carried out in the picture fuzzy environment. Hence, we have presented mining safety model system in terms of picture fuzzy differential equations. We have discussed both analytically and numerically the solution of the model system in details. Moreover, outcomes of our model systems are compared using classical method, Zadeh's extension principle, and generalized Hukuhara differentiability (GH_{ii}).

1.5. Structure of this paper

The remaining portions of this article are ornamented as follows: Section 2 gives some preliminary concepts and definitions associated with picture fuzzy numbers. Section 3 contains first Order linear homogeneous picture fuzzy differential equation. Section 4 contains description of MSMS. Section 5 contains mathematical formulation of MSMS. Section 6 contains analytical solution of the MSMS model system Section 7 contains numerical solution of the MSMS model system. Finally, conclusion are discussed in Section 8.

2. Preliminaries

Definition 2.1 (*[19]*). Let U be a universal set. A PFS A^{PS} of U is defined by $A^{PS} = \hat{e}(u; {}_A(u), {}_A(u); {}_A(u)) : u \in U\ddot{e}$, where ${}_A(u); {}_A(u); {}_A(u)$ is outlined as the positive membership (PM), neutral membership (Neum), and negative membership (NM) grades of u in A^{PS}, which are subsets of [0; 1] and ${}_A(u) + {}_A(u) + {}_A(u) \acute{I} 1$.

Definition 2.2. We conclusions TPFNs in a modified form as defined by Dyivya et al. [42] in the following manners:

A TPFNs is denoted by $A^{PS} = \hat{e}a'_1; a'_2; a'_3; w\ ; w_{\#}; w\ \ddot{e}$, whose positive (${}_A(u)$), neutral (${}_A(u)$), and negative (${}_A(u)$) membership functions are respectively defined as follows:

$$ {}_A(u) = \begin{cases} n\left(\frac{u-a'_1}{a'_2-a'_1}\right)w & \text{when } a'_1 \acute{I} u \acute{I} a'_2 \\ nw & \text{when } u = a'_2 \\ n\left(\frac{a'_3-u}{a'_3-a'_2}\right)w & \text{when } a'_2 \acute{I} u \acute{I} a'_3 \\ n0 & \text{whenu } \acute{I} a'_1 \text{ or } u \hat{I} a'_3 \end{cases} $$

$$ {}_A(u) = \begin{cases} n\frac{(a'_2-u)+(u-a'_1)w_{\#}}{a'_2-a'_1} & \text{when } a'_1 \acute{I} u \acute{I} a'_2 \\ nw_{\#} & \text{when } u = a'_2 \\ n\frac{(u-a'_2)+(a'_3-u)w_{\#}}{a'_3-a'_2} & \text{when } a'_2 \acute{I} u \acute{I} a'_3 \\ n1 & \text{when } u \acute{I} a'_1 \text{ or } u \hat{I} a'_3 \end{cases} $$

$$ {}_A(u) = \begin{cases} n\frac{(a'_2-u)+(u-a'_1)w}{a'_2-a'_1} & \text{when } a'_1 \acute{I} u \acute{I} a'_2 \\ nw & \text{when } u = a'_2 \\ n\frac{(u-a'_2)+(a'_3-u)w}{a'_3-a'_2} & \text{whena}'_2 \acute{I} u \acute{I} a'_3 \\ n1 & \text{whenu } \acute{I} a'_1 \text{ or } u \hat{I} a'_3 \end{cases} $$

where ${}_A(u) + {}_A(u) + {}_A(u) \acute{I} 1 \& w \in (0; 1], w_{\#}; w \in [0; 1)$.

Definition 2.3. We defined the $\hat{e}\ ;\ ;\ \ddot{e}$-cut of PFS in a modified form from that defined by Dyivya et al. [42] in the following manners:

Let A^{PS} be any PFS, then the $\hat{e}\ ;\ ;\ \ddot{e}$-cut of PFS is denoted by $A^{PS}(\ ;\ ;\ )$, and it is defined by $A^{PS}(\ ;\ ;\ ) = \hat{e}v \in V :\ {}_A(v) \hat{I}\ ;\ {}_A(v) \acute{I}\ ;\ {}_A(v) \acute{I}\ ; 0 < \acute{I} 1; 0 \acute{I} < 1; 0 \acute{I} < 1\ddot{e}$.

Now the basic arithmetic operation on TPFNs can be written as:

Consider two TPFNs, $A^{PS} = \hat{e}a_1; a_2; a_3; w\ ; w_{\#}; w\ \ddot{e}$; $B^{PS} = \hat{e}b_1; b_2; b_3; u\ ; u_{\#}; u\ \ddot{e}$. Then the addition, subtraction, multiplication, and division of two TPFNs can be defined as follows:

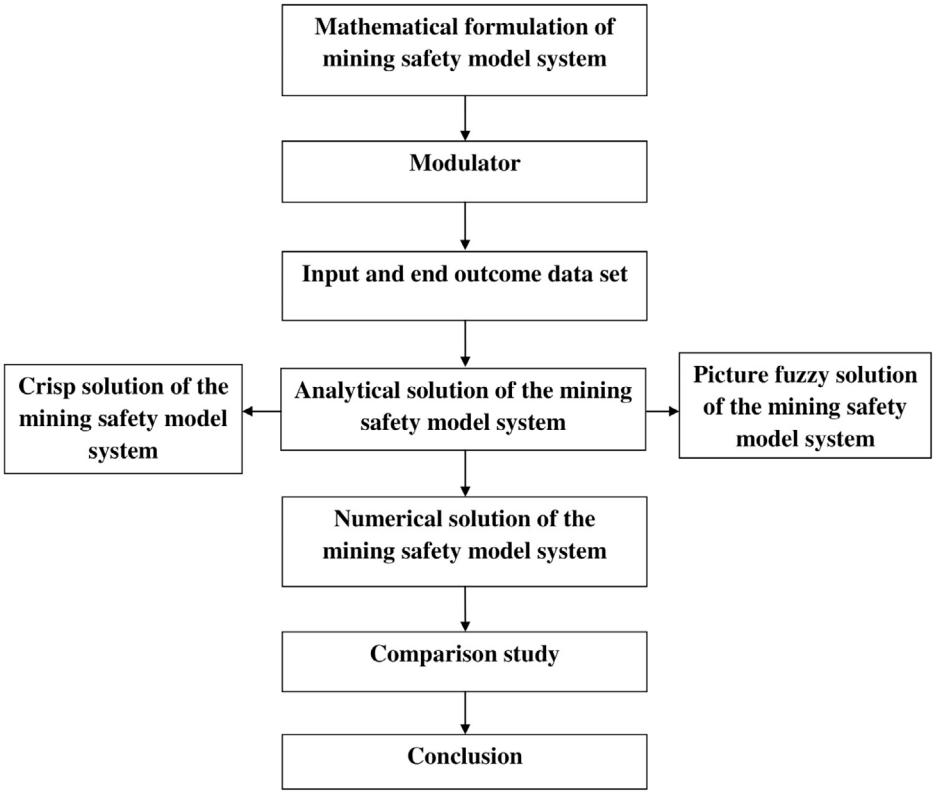

Fig. 1. Flowchart of the proposed article.

Addition of two TPFNs : $A^{PS}+B^{PS} = \hat{e}[(a_1+b_1; a_2+b_2; a_3+b_3); w \wedge u; w_{\#} \vee u_{\#}; w \vee u]ë$

Subtraction of two TPFNs: $A^{PS}-B^{PS} = \hat{e}[(a_1-b_3; a_2-b_2; a_3-b_1); w \wedge u; w_{\#} \vee u_{\#}; w \vee u]ë$

Multiplication of two TPFNs: $A^{PS}:B^{PS} = \hat{e}[(a_1b_1; a_2b_2; a_3b_3); w \wedge u; w_{\#} \vee u_{\#}; w \vee u]ë$

Division of two TPFNs: $\frac{A^{PS}}{B^{PS}} = \hat{e}[(\frac{a_1}{b_3}; \frac{a_2}{b_2}; \frac{a_3}{b_1}); w \wedge u; w_{\#} \vee u_{\#}; w \vee u]ë$, where $\wedge = Min; \vee = Max$.

3. First order linear homogeneous picture fuzzy differential equation

[44] Let us consider an ODE

$$\frac{dY}{dt} = KY; t \in [0; \infty) \quad (1)$$

with IC $Y(t_0) = Y_0$. Now the above ODE will be treated as PFDE when at least one of the parameter K or initial condition Y_0 are considered as PFNs.

Now, the classical technique [45], Zadeh's extension principle [46], generalized Hukuhara differentiability (GH_{ii}) [47], and other approaches can each be used to solve the PFDE mentioned above.

3.1. Definition of strong solution

Let $\dot{Y}^{PS}(t)$ be the classical solution (as described in [45]) and its Cut be $Y(t; ; ;) = \hat{e}[Y_1(t;), Y_2(t;)]; [Y_1'(t;); Y_2'(t;)]; [Y_1''(t;); Y_2''(t;)]ë$. The solution is strong if

(i) $\frac{dY_1(t;)}{d} > 0, \frac{dY_2(t;)}{d} < 0 \ \forall \in (0; 1], Y_1(t; 1) Í Y_2(t; 1).$

(ii) $\frac{dY_1'(t;)}{d} < 0, \frac{dY_2'(t;)}{d} > 0 \ \forall \in [0; 1), Y_1'(t; 0) Í Y_2'(t; 0).$

(iii) $\frac{dY_1''(t;)}{d} < 0, \frac{dY_2''(t;)}{d} > 0 \ \forall \in [0; 1), Y_1''(t; 0) Í Y_2''(t; 0).$

A weak solution would result from anything less (see Fig. 1).

4. Description of MSMS

The development of human civilization has been significantly influenced by the mining industry. There are several dangers associated with mining minerals from subterranean systems of work, including roof collapse, workplace flooding from water incursions from previous operations, explosions, an increase in dangerous gas levels, etc. Similar risks have been associated with the opencast system of labor, including the possibility of dumper runways, worksite bench slides, rock blasting strikes, etc. This problem not only harms the workers, but it can also be lethal in rare cases. Accidents can also be caused by improper usage of mining equipment or systems. Here, we proposed to study a MSMS in terms of PFDE. Our main interests in this paper are given below:

- Establish a model for MSMS.
- Determine the model's solution in a CS environment.
- Determine the model's solution in the PFS environment.

5. Mathematical formulation of MSMS

5.1. Acceptation

(I) Every event exists independently of the others.

(II) In the finite time period t, the probability of changing from one system state to another is given by the expression t, where is the rate of change between system states.

(III) $(t)(t) ™ 0.$

(IV) $P\{ (t) Î 2\} = o(t)$, where (t) be the number of event that occur in t.

(V) $P\{ (t) = 1\} = t + o(t)$, where > 0.

(VI) $\lim_{t™0} \frac{o(t)}{t} = 0.$

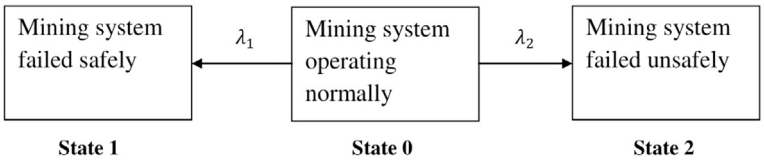

Fig. 2. The mining system state space diagram depicts the rate of change from the mining system operating normally (state 0) to the mining system safe failure state (state 1) and mining system unsafe failure state (state 2).

5.2. Modulator

t	Time.
t	Finite time interval.
$_1$	Rate of failure of the mining system under safe condition.
$_2$	Rate of failure of the mining system under unsafe condition.
$P_0(t+\ t)$	Operating probability in position 0 at time t + t.
$P_1(t+\ t)$	Probability of a safe failure in position 1 at time t + t.
$P_2(t+\ t)$	Probability of a unsafe failure in position 2 at time t + t.
$j = 0$	Position in normal.
$j = 1$	Position fail safe.
$j = 2$	Position fail unsafe.
$P_j(t)$	Probability in position j at time t.
$_1\ t$	Probability of a safe failure in t
$_2\ t$	Probability of a unsafe failure in t.
$(1 - _1\ t)$	No safe fail probability in t.
$(1 - _2\ t)$	No unsafe fail probability in t.

5.3. Input data

t = Time.
$_1$ = MSMS-Safe-Failure-Rate.
$_2$ = MSMS-UnSafe-Failure-Rate.

5.4. The end outcome

$P_0(t)$ = Probability of Mining system operating normally.
$P_1(t)$ = Probability of Mining system failed safely.
$P_2(t)$ = Probability of Mining system failed unsafely.

An MSMS can be shut down safely or unsafely. Fig. 2 depicts the mining system's state space diagram. The numbers 0, 1 and 2 in the boxes represent the statuses of the MSMS. State 0 indicates MSMS operating normally, and State 1 indicates mining system shutdown safely, and State 2 indicates MSMS shutdown unsafely. Let $_1$ indicate the rate of failure safely from state 0 to state 1 and $_2$ indicate the rate of failure unsafely from state 0 to state 2. The meaning of t, t, $P_0(t+\ t)$, $P_1(t+\ t)$, $P_2(t+\ t)$, $P_0(t)$, $P_1(t)$, $P_2(t)$, $_1\ t$, $_2\ t$, $(1 - _1\ t)$ and $(1 - _2\ t)$ already discussed in Section 5.2. Assume that the MSMS is based on the suppositions that system failures happen independently and that the rates of system failure in both the shutdown safely and shutdown unsafely modes are constant.

Now using the notation of Section 5.1, it is clear that at the initial time, $P_0(0) = 1$, $P_1(0) = 0$ and $P_2(0) = 0$. Now using the concept of basic probability theory, the following three equations can be written:

$$P_0(t+\ t) = P_0(t)(1 - _1\ t)(1 - _2\ t) \tag{2}$$

$$P_1(t+\ t) = P_1(t)(1 - o(\ t)) + P_0(t)\ _1\ t \tag{3}$$

$$P_2(t+\ t) = P_2(t)(1 - o(\ t)) + P_0(t)\ _2\ t \tag{4}$$

Now, as $(_1\ t)(_2\ t) \to 0$, so from (2) we get

$$P_0(t+\ t) - P_0(t) = -(_1 + _2)P_0(t)\ t \tag{5}$$

Now using definition of differentiability from Eq. (5), we get

$$\lim_{t \to 0} \frac{P_0(t+\ t) - P_0(t)}{t} = -(_1 + _2)P_0(t)$$

$$\therefore \frac{dP_0(t)}{dt} = -(_1 + _2)P_0(t) \tag{6}$$

Similarly from Eqs. (3) and (4), we get

$$\frac{dP_1(t)}{dt} = _1P_0(t) \tag{7}$$

$$\frac{dP_2(t)}{dt} = _2P_0(t) \tag{8}$$

So, the required MSMS model system as follows

$$\frac{dP_0(t)}{dt} = -(_1 + _2)P_0(t) \tag{9}$$

$$\frac{dP_1(t)}{dt} = _1P_0(t) \tag{10}$$

$$\frac{dP_2(t)}{dt} = _2P_0(t) \tag{11}$$

with initial conditions: $P_j(0) = 1$ for $j = 0$ & $P_j(0) = 0$ for $j = 1,2$.

6. Analytical solution of the MSMS model system

The Eqs. (9), (10) and (11) given above have been solved in two different environments, namely in a CS environment and an imprecise environment. We have chosen the underlying model parameters as PFS numbers in an imprecise environment. The details of solutions are discussed below:

6.1. CS solution of the MSMS model system:

Taking both the input values $_1$ and $_2$ as crisp numbers, we get the following solutions:

$$P_0(t) = e^{-(_1 + _2)t} \tag{12}$$

$$P_1(t) = \frac{_1}{_1 + _2}[1 - e^{-(_1 + _2)t}] \tag{13}$$

$$P_2(t) = \frac{_2}{_1 + _2}[1 - e^{-(_1 + _2)t}] \tag{14}$$

6.2. PFS solution of the MSMS model system:

The following combinations of input values are used to arrive at the PFS solution for the system of Eqs. (9), (10), and (11):
Case-1: When f_1^{PS} is PFNs & $_2$ is CSNs.
Case-2: When $_1$ is CSNs & f_2^{PS} is PFNs.
Case-3: When both f_1^{PS} & f_2^{PS} are PFNs.

In the above cases let, $P_0(t)^{PS}$, $P_1(t)^{PS}$, $P_2(t)^{PS}$ be the PFS solution of the model system which have the following ; ; -cut values:

$$P_0(t;\ ;\ ;\ ) = \hat{e}[P_{01}(t;\ );P_{02}(t;\ )];[P'_{01}(t;\ );P'_{02}(t;\ )];[P''_{01}(t;\ );P''_{02}(t;\ )]\ddot{e}$$

$$P_1(t;\ ;\ ;\ ) = \hat{e}[P_{11}(t;\ );P_{12}(t;\ )];[P'_{11}(t;\ );P'_{12}(t;\ )];[P''_{11}(t;\ );P''_{12}(t;\ )]\ddot{e}$$

$$P_2(t;\ ;\ ;\ ) = \hat{e}[P_{21}(t;\ );P_{22}(t;\ )];[P'_{21}(t;\ );P'_{22}(t;\ )];[P''_{21}(t;\ );P''_{22}(t;\ )]\ddot{e}$$

Table 2
f_1^{PS} is TPFN and $_2$ is crisp number.

; ;	$P_{11}(t;\ )$	$P_{12}(t;\ )$	$P'_{11}(t;\ )$	$P'_{12}(t;\ )$	$P''_{11}(t;\ )$	$P''_{12}(t;\ )$	$P_{21}(t;\ )$	$P_{22}(t;\ )$	$P'_{21}(t;\ )$	$P'_{22}(t;\ )$	$P''_{21}(t;\ )$	$P''_{22}(t;\ )$
0	0.124044	0.203754	0.167581	0.158722	0.173156	0.153223	0.017721	0.018523	0.018172	0.018082	0.018227	0.018026
0.1	0.130452	0.196883	0.163142	0.163142	0.168137	0.158171	0.017789	0.018458	0.018127	0.018127	0.018177	0.018077
0.2	0.136905	0.190053	0.158722	0.167581	0.163142	0.163142	0.017857	0.018392	0.018082	0.018172	0.018127	0.018127
0.3	0.143400	0.183263	0.154321	0.172039	0.158171	0.168137	0.017925	0.018326	0.018038	0.018216	0.018077	0.018177
0.4	0.149938	0.176515	0.149938	0.176515	0.153223	0.173156	0.017993	0.018260	0.017993	0.018260	0.018026	0.018227
0.5	0.156519	0.169808	0.145575	0.181009	0.148300	0.178198	0.018060	0.018194	0.017948	0.018304	0.017976	0.018277
0.6	0.163142	0.163142	0.141230	0.185522	0.143400	0.183263	0.018127	0.018127	0.017902	0.018348	0.017925	0.018326
0.7	0.169808	0.156519	0.136905	0.190053	0.138524	0.188352	0.018194	0.018060	0.017857	0.018392	0.017874	0.018376
0.8	0.176515	0.149938	0.132598	0.194602	0.133673	0.193463	0.018260	0.017993	0.017812	0.018436	0.017823	0.018425
0.9	0.183263	0.143400	0.128311	0.199169	0.128846	0.198597	0.018326	0.017925	0.017766	0.018480	0.017772	0.018474
1.0	0.190053	0.136905	0.124044	0.203754	0.124044	0.203754	0.018392	0.017857	0.017721	0.018523	0.017721	0.018523

7. Numerical solution of the MSMS model system

Here, we have assumed 0.009 and 0.001 are the rates of failures per hour in safe and unsafe mode respectively (data reference: [43]). In the following sub-section, we have estimated the probability of 20 h expedition's MSMS failing due to safe and unsafe failures in (i) the CS environment and (ii) the PFS environment.

7.1. CS solution

When we take the respective input values $_1 = 0{:}009$, $_2 = 0{:}001$, and t = 20 − h, we get the following output results: $P_1(20) = 0.163142$ and $P_2(20) = 0.018127$.

7.2. PFS solution

In practise, the safe failure rate $_1$ and unsafe failure rate $_2$ in mine safety management are continually changing owing to a variety of external and internal causes. When solving the mining model with these variable values, various ambiguities and uncertainties can be observed. In this situation, the results obtained by solving the model assuming safe failure and unsafe failure rates as picture fuzzy numbers seem to be very acceptable. We numerically solve the model system in the picture fuzzy environment in the following three cases:

7.2.1. **Case: 1** *(When f_1^{PS} is TPFNs and $_2$ is CSNs)*

In this case, the parameter $_1$ is assumed to be a picture-fuzzy number, and the parameter $_2$ remains unchanged. When the parameter $_1$ is changed from a crisp to a picture-fuzzy number, the maximum tolerance to its left and right has been extended up to 22.2%. Here, the maximum positive membership value is assumed to be 0.6, with the minimum neutral and negative membership values being 0.1 and 0.2, respectively.

We choose $f_1^{PS} = \hat{e}0{:}007; 0{:}009; 0{:}011; 0{:}6; 0{:}1; 0{:}2\ddot{e}$; $_2 = 0{:}001$ and t = 20 − h in this case. Table 2 shows the numerical values of the $\hat{e}$; ; $\ddot{e}$-cut of the probability of MSMS failing safely $[P_1(t;\ ;\ ;\ )]$ and the probability of MSMS failing unsafely $[P_2(t;\ ;\ ;\ )]$ for $\in (0; 1]$ and ; $\in [0; 1)$. Table 2 shows that when increases, the values of $P_{11}(t;\ )$ and $P_{21}(t;\ )$ increase while the values of $P_{12}(t;\ )$ and $P_{22}(t;\ )$ decrease. If both and increase, the values of $P'_{12}(t;\ )$, $P'_{22}(t;\ )$, $P''_{12}(t;\ )$, and $P''_{22}(t;\ )$ increase, while $P'_{11}(t;\ )$, $P'_{21}(t;\ )$, $P''_{11}(t;\ )$, and $P''_{21}(t;\ )$ decrease. We created a graph of the found solution, which is displayed in Figs. 3 and 4. Fig. 3 shows the graph of the mining system's positive, neutral, and negative membership functions when it is shut down in safe mode, and Fig. 4 shows the graph of the mining system's positive, neutral, and negative membership functions when it is shut down in unsafe mode. So the triangular picture fuzzy value of the probability that MSMS fails safely in a 20 h time interval is $P_1(20)^{PS} = \hat{e}0{:}124044; 0{:}163142; 0{:}203704; 0{:}6; 0{:}1; 0{:}2\ddot{e}$, where the positive ($_{P_1}(u)$), neutral ($_{P_1}(u)$), and negative ($_{P_1}(u)$) membership functions are as follows:

$$_{P_1}(u) = \begin{cases} (\frac{u-0:124044}{0:039098}):0{:}6 & \text{when } 0{:}124044 \acute{I} u \acute{I} 0{:}163142 \\ 0{:}6 & \text{when } u = 0{:}163142 \\ (\frac{0:203704-u}{0:040562}):0{:}6 & \text{when } 0{:}163142 \acute{I} u \acute{I} 0{:}203704 \\ 0 & \text{whenu } \acute{I} 0{:}124044 \text{ or } u \hat{I} 0{:}203704 \end{cases}$$

$$_{P_1}(u) = \begin{cases} \frac{0:1507376-0:9:u}{0:039098} & \text{when } 0{:}124044 \acute{I} u \acute{I} 0{:}163142 \\ 0{:}1 & \text{when } u = 0{:}163142 \\ \frac{0:9u-0:1427716}{0:040562} & \text{when } 0{:}163142 \acute{I} u \acute{I} 0{:}203704 \\ 1 & \text{whenu } \acute{I} 0{:}163142 \text{ or } u \hat{I} 0{:}203704 \end{cases}$$

$$_{P_1}(u) = \begin{cases} \frac{0:1383332-0:8:u}{0:039098} & \text{when } 0{:}124044 \acute{I} u \acute{I} 0{:}163142 \\ 0{:}2 & \text{when } u = 0{:}163142 \\ \frac{0:8u-0:1224012}{0:040562} & \text{when } 0{:}163142 \acute{I} u \acute{I} 0{:}203704 \\ 1 & \text{when } u \acute{I} 0{:}163142 \text{ or } u \hat{I} 0{:}203704 \end{cases}$$

where $_{P_1}(u) + _{P_1}(u) + _{P_1}(u) \acute{I} 1$

Also, the triangular picture fuzzy value of the probability that MSMS failed in an unsafe manner during a 20 h mission is $P_2(20)^{PS} = \hat{e}0{:}017721; 0{:}018127; 0{:}018523; 0{:}6; 0{:}1; 0{:}2\ddot{e}$ with positive ($_{P_2}(u)$), neutral ($_{P_2}(u)$), and negative ($_{P_2}(u)$) membership functions as follows:

$$_{P_2}(u) = \begin{cases} (\frac{u-0:017721}{0:000406}):0{:}6 & \text{when } 0{:}017721 \acute{I} u \acute{I} 0{:}018127 \\ 0{:}6 & \text{when } u = 0{:}018127 \\ (\frac{0:018523-u}{0:000396}):0{:}6 & \text{when } 0{:}018127 \acute{I} u \acute{I} 0{:}018523 \\ 0 & \text{when } u \acute{I} 0{:}017721 \text{ or } u \hat{I} 0{:}018523 \end{cases}$$

$$_{P_2}(u) = \begin{cases} \frac{0:0163549-0:9:u}{0:000406} & \text{when } 0{:}017721 \acute{I} u \acute{I} 0{:}018127 \\ 0{:}1 & \text{when } u = 0{:}018127 \\ \frac{0:9u-0:0162747}{0:000396} & \text{when } 0{:}018127 \acute{I} u \acute{I} 0{:}018523 \\ 1 & \text{when } u \acute{I} 0{:}0{:}017721 \text{ or } u \hat{I} 0{:}018523 \end{cases}$$

$$_{P_2}(u) = \begin{cases} \frac{0:0145828-0:8:u}{0:000406} & \text{when } 0{:}017721 \acute{I} u \acute{I} 0{:}018127 \\ 0{:}2 & \text{when } u = 0{:}018127 \\ \frac{0:8u-0:0144224}{0:000396} & \text{when } 0{:}018127 \acute{I} u \acute{I} 0{:}018523 \\ 1 & \text{when } u \acute{I} 0{:}0{:}017721 \text{ or } u \hat{I} 0{:}018523 \end{cases}$$

where $_{P_2}(u) + _{P_2}(u) + _{P_2}(u) \acute{I} 1$.

In the MSMS system, with a time interval of 20 h, the crisp values of the safe and unsafe failure probabilities are 0.163142 and 0.018127, respectively. As well as for the same time interval, the triangular picture fuzzy values of the safe and unsafe failure probabilities are $\hat{e}0{:}124044; 0{:}163142; 0{:}203704; 0{:}6; 0{:}1; 0{:}2\ddot{e}$ and $\hat{e}0{:}017721; 0{:}018127; 0{:}018523; 0{:}6; 0{:}1; 0{:}2\ddot{e}$, respectively. Here the values in the picture fuzzy environment are much more flexible when compared to crisp environment. As well as in the picture fuzzy environment of the MSMS system, the calculated safe failure probability values indicate that its left and right maximum allowable tolerances are 24% and 25%, respectively, the maximum positive membership value is 0.6, the minimum neutral and negative membership values are 0.1 and 0.2, respectively. Again,